

DGCA EXAMINATION PRACTICE PAPER

1. A Rhumb Line cuts all meridians at the same angle. This gives:-

- a) The shortest distance between two points.
- b) A line which could never be a great circle track
- c) A line of constant direction

2. What is the dip angle at the South Magnetic Pole?

- a) 0°
- b) 90°
- c) 180°
- d) 64°

3. If variation is West, then:

- a) True North is West of Magnetic North
- b) Compass North is West of Magnetic North
- c) True North is East of Magnetic North
- d) Magnetic North is West of Compass North

4. An airfield has runways 18/36 and 07/25, all of equal length. The TAF W/V is 310/25. Variation is 20°E . Which will be the best R/W for take-off and landing?

- a) 18
- b) 36
- c) 07
- d) 25

5. On the approach to London Heathrow runway 27, glide slope 3°, you reduce speed from 150 kts to 120 kts. What change should you make to your ROD to maintain glideslope?
6. A flight is to be made along the parallel of latitude from A at 48°00'N 04°00'W to B at 48°00'N 02°27'E. Calculate the distance.
7. An aircraft flies for 1000nm along a rhumb line track of 090°(T) from C at 36°00'N 174°45'E to D. What is the longitude of D?
8. Your position is 5833N 17400W. You fly exactly 6 nm eastwards. What is your new position?
 - a) 5833N 17411.5W
 - b) 5833N 17355W
 - c) 5833N 17340W
 - d) 5833N 17348.5W
9. On a chart, 49 nautical miles is represented by 7.0 centimetres. What is the scale?
 - a) 1 : 700,000
 - b) 1 : 2,015,396
 - c) 1 : 1,296,400
 - d) 1 : 1,156,600
10. On a particular Direct Mercator wall chart, the 180°W to 180°E parallel of latitude at 53° N is 133 cm long. What is the scale of the chart at 30°S?
 - a) 1: 3,000,000
 - b) 1: 18,000,000
 - c) 1: 21,000,000
 - d) 1: 27,000,000
11. A blocked pitot probe will affect?
 - a. ASI.
 - b. VSI.
 - c. Altimeter.
 - d. All of the above.

12. The hysteretic error of an altimeter varies substantially with the?

- a. Time passed at a given altitude.
- b. Mach number of the aircraft.
- c. Aircraft altitude.
- d. Static temperature.

13. If ONH is 999 hPa, what is the pressure altitude at an elevation of 25000 ft?

- a. 25100ft.
- b. 25200 ft.
- c. 25300 ft
- d. 25400 ft.

14. At 60S on a Mercator chart, chart convergence is:

- a) greater than Earth convergency
- b) "correct"
- c) less than Earth convergency
- d) equal to $ch \text{ long} \times 0.866$

15. On a Lambert's chart the constant of the cone is 0.80. A is at 53N 04W. You plan to fly to B. The initial Lambert's chart straight-line track is $070^\circ(T)$ and the Rhumb Line track from A to B is $082^\circ(T)$. What is the longitude of B?

- a) 26E
- b) 34W
- c) 11E
- d) 15E

16. SAT = $+35^\circ C$. Pressure alt (corrected for barometric error) = 5000 feet. What is true alt?

(2 Marks)

- a) 4550 feet
- b) 5550 feet
- c) 4920 feet
- d) 5320 feet

17. What is the approximate compression of the Earth?

- a) 3%
- b) 0.03%
- c) 0.3%
- d) 1/3000

18. You plan to fly a track (course) of 348° (T), Drift is 17° port, Variation = 32° W. Deviation is 4° E. What compass heading should you fly?

(2 Marks)

- a) 041°
- b) 033°
- c) 016°
- d) 359°

19. At a specific location, the value of magnetic variation

(1 Mark)

- a) depends on the value of magnetic heading
- b) depends on the value of true heading
- c) varies slowly over time
- d) depends on the type of compass installed

20. Given:

True Track = 352° (T)

Variation = 11° W

Deviation = -5°

Drift = 10° R

What is Heading $^\circ$ (C)?

- a) 078° (C)
- b) 346° (C)
- c) 358° (C)
- d) 025° (C)

21. You are flying at FL330 at Mach No 0.9M. Ambient temperature is ISA +15°. What is your TAS?

(1 Mark)

- a) 600 knots
- b) 595 knots
- c) 540 knots
- d) 505 knots

22. You plan to land on R W 14. The met forecast wind velocity is 110 30. Variation is 30°W. What crosswind do you expect?

(1 Mark)

- a) 15 kts
- b) 0 kts
- c) 26 kts
- d) 30 kts

23. An aircraft is maintaining a 5.2% gradient on a flat terrain. Its height at 7 nm from the runway is approximately?

(2 Marks)

- a) 3640 feet
- b) 1890 feet
- c) 2210 feet
- d) 680 feet

24. Course 040°(T), TAS 120 kn, Wind speed = 30 knots. From which direction will the wind give the greatest drift?

(1 Mark)

- a) 215°(T)
- b) 230°(T)
- c) 235°(T)
- d) 240°(T)

25. An aircraft is climbing at a constant CAS in ISA conditions. What will be the effect on TAS and Mach No?

(1 Mark)

- a) TAS increases and Mach No decreases
- b) Both increase
- c) Both decrease
- d) TAS decreases and Mach No increases

26. Convert 70 metres/sec into knots.

(1 Mark)

- a) 136 knots
- b) 36 knots
- c) 146 knots
- d) 54 knots

Q.27 What is the ratio between the litre and the US-GAL? 1 US-GAL equals?

- a. 3.5 litres
- b. 4.3 litres
- c. 3.78 litres
- d. 3.65 litres

Q.28 Given: true track is 348° , drift 17° left, variation 32° W, deviation 4° E. What is the compass heading?

- a. 041°
- b. 141°
- c. 044°
- d. 033°

Q.29 An aircraft travels 100 statute miles in 20 min, how long does it take to travel 215 nm?

- a. 50 min
- b. 46 min
- c. 48 min
- d. 51 min

30. Airfield elevation is 1000 feet. The QNH is 988. Use 27 feet per hectopascal. What is pressure altitude?

(1 Mark)

- a) 675 feet
- b) 325 feet
- c) 1675 feet
- d) 825 feet

31. A non-perspective chart:

(1 Mark)

- a) is produced directly from a light projection of a Reduced Earth
- b) cannot be used for navigation
- c) is produced by mathematically adjusting a light projection of the Reduced Earth
- d) is used for a Polar Stereographic projection

32. The maximum difference between Mean Time and Apparent Time is:

- a) 21 minutes
- b) 16 minutes
- c) 30 minutes
- d) there is no difference

33. What is the length of a Sidereal Year?

- a) 365 days
- b) 366 days
- c) 365 days 6 hrs
- d) 365 days 5 hrs 48.75 minutes

34. Given: TAS 140 kt, HDG 005°(T), W/V 265/25, calculate the drift and groundspeed.

- a) 11R - 140 kt
- b) 10R - 146 kt
- c) 9R - 140 kt
- d) 11R - 142 kt

(2 marks)

35. Given: FL 120, OAT is ISA, CAS = 200 knots, Track = 222°(M), Heading = 215°(M), Variation = 15°W.

If the time to fly 105 nm is 21 minutes, what is the W/V?

- a) 040T / 105 kt
- d) 055T / 105 kt
- c) 050T / 70 kt
- d) 065T / 70 kt

(2 marks)

36. Fuel flow per HR is 22 US-GAL, total fuel on board is 83 IMP GAL. What is the endurance ?

- a. 4 hr 32 min
- b. 3 hr 46 min
- c. 4 hr 46 min
- d. 3 hr 32 min

37. Given: True track 180° Drift 8°R Compass heading 195° Deviation -2° Calculate the variation?

- a. 23°W
- b. 20°W
- c. 21°W
- d. 21°W.

39. Given: true course A to B = 250° Distance A to B= 315 NM TAS=450 kt. W/V=200°/60kt. ETD A= 0650 UTC. What is the ETA at B?

- a. 0726 UTC.
- b. 0659 UTC.
- c. 0732 UTC.
- d. 0736 UTC.

40. Given: Runway direction 083° (M), Surface W/V 035/35kt. Calculate the effective headwind component?

- a. 24 kt.
- b. 20 kt.
- c. 22 kt.
- d. 26 kt

41. On a Lambert conformal conic, the distance between parallels of latitude spaced the same number of degrees apart?

- a. Increases between and reduce outside of the standard parallels.
- b. Reduces between, and expands outside, the standard parallels.
- c. Is constant between the standard parallels.
- d. Is constant outside of the standard parallels.

42. The 'departure' between positions 60°N 60°E and 60° N 'x' is 1800 NM East. What is the longitude of 'x'?

- a. 145° E.
- b. 145° W.
- c. 120° E.
- d. 120° W.

43. A straight line on a Lambert Conformal Projection chart for normal flight planning purpose?

- a. Is a Rhumb line.
- b. Can only be a parallel of latitude.
- c. Is approximately a Great Circle.
- d. Is a Loxodromic line.

44. Isogonals converge at the?

- a. Magnetic equator.
- b. North magnetic pole only.
- c. North and South magnetic poles only.
- d. North and South geographic and magnetic poles.

45. What is the angle between True North and Magnetic North?

- a. Drift.
- b. Variation.
- c. Dip.
- d. Deviation.

46. revenue flight is to be made by a jet transport. The following are the aeroplane's structural limits:-Maximum Ramp Mass: 69 900 kg-Maximum Take Off Mass: 69 300 kg-Maximum Landing Mass: 58 900 kg-Maximum Zero Fuel Mass: 52 740 Kg Take Off and Landing mass are not performance limited .Dry Operating Mass: 34 930 kg Trip Fuel: 11 500 kg Taxi Fuel: 250 kg Contingency & final reserve fuel: 1 450 Kg Alternate Fuel: 1 350 kg The maximum traffic load that can be carried is:

- a) 21 170 kg
- b) 17 810 kg
- c) 21 070 kg
- d) 20 420 kg

47. With respect to aeroplane loading in the planning phase, which of the following statements is always correct ?

- a. $MTOM = ZFM + \text{maximum possible fuel mass}$
- b. $MZFM = \text{Traffic load} + DOM$
- c. $LM = TOM - \text{Trip Fuel}$
- d. $\text{Reserve Fuel} = TOM - \text{Trip Fuel}$

48. 730 FT/MIN equals?

- a. 5.2 m/sec.
- b. 1.6 m/sec.
- c. 2.2 m/sec.
- d. 3.7 m/sec.

49. If there is a 15 knot increase in headwind by what amount must the rate of descent be changed in order to maintain a 3° glideslope?

- a. It must be decreased by 79 ft/ min.
- b. It must be increased by 79 ft/ min.
- c. It must be increased by 35 ft/ min.
- d. It must be decreased by 35 ft/ min.

50. At 0422 an aircraft at FL370, GS 320kt, is on the direct track to VOR 'X' 185 NM distant. The aircraft is required to cross VOR 'X' at FL80. For a mean rate of descent of 1800 FT/MIN at a mean GS of 232 kt, the latest time at which to commence descent is?

- a. 0451.
- b. 0454.
- c. 0445.
- d. 0448.

51. An aircraft was over 'Q' at 1320 hours flying direct to 'R'. Given : Distance 'Q' to 'R' 3016 NM, True airspeed 480 kt, Mean wind component 'out' -90 kt, Mean wind component 'back' +75 kt, Safe endurance 10:00 HR . The distance from 'Q' to the Point of Safe Return (PSR) 'R' is?

- a. 1510 NM.
- b. 2290 NM.
- c. 2370 NM.
- d. 1310 NM.

52. How far can the aeroplane fly out from its base and return in one hour, when flying at TAS 260 kt on a track of 090°, if the W/V is 045°/50kt?
- a. 158 nm.
 - b. 128 nm.
 - c. 188 nm.
 - d. 175 nm.
53. What is the chart distance between longitudes 179° E and 175°W on a direct Mercator chart with a scale of 1 : 5 000 000 at the equator?
- a. 133 mm.
 - b. 100 mm.
 - c. 113 cm.
 - d. 125 mm.
54. What is the meaning of the term “standard time”?
- a. It is the time set by the legal authorities for a country or part of country.
 - b. It is just another term for UTC.
 - c. It is time related to the Prime meridian.
 - d. It is constant at all points on the earth.
55. What is the local mean time, position 65°25'N 123°45'W at 2200 UTC?
- a. 0615.
 - b. 1815.
 - c. 1345.
 - d. 1415.
56. The main reason that day and night, throughout the year, have different duration, is due to the?
- a. The changing distance between the Sun and the Earth.
 - b. The changing speed of rotation of the earth.
 - c. The changing shape of the orbit to the Earth around the Sun.
 - d. The inclination of the ecliptic to the Equator.
57. If QNH is 1000 hPa and field elevation is 4500 ft amsl, what is QFE?
- a. 850 hPa.
 - b. 163 hPa.
 - c. -850 hPa.
 - d. 900 hPa.

58. Pressure altitude is?

- a. The altitude above sea level.
- b. The altimeter indication when QFE is set on the sub-scale.
- c. The altimeter indication when QNH is set on the subscale.
- d. The altimeter indication when 1013.25 hPa is set on the sub-scale.

59. Which of the following cause air density to decrease?

- a. Increasing humidity, increasing altitude, increasing temperature.
- b. Increasing humidity, increasing altitude, decreasing temperature.
- c. Increasing humidity, decreasing altitude, increasing temperature.
- d. Decreasing humidity, increasing altitude, decreasing temperature.

60. If QNH changes from 1013 hPa to 1022 hPa this will?

- a. Increase field elevation.
- b. Decrease field elevation.
- c. Not affect field elevation.
- d. Decrease QFE.

61. What is the true altitude of an aircraft flying at 16000 ft indicated altitude with an OAT of -16 degrees C?

- a. 13200ft.
- b. 14200 ft.
- c. 16050ft.
- d. 16200ft.

62. From where does the ADC obtain altitude data?

- a. Radio Altimeter.
- b. OAT sources.
- c. Barometric altitude source.
- d. Dynamic minus total pressure.

63. When descending through an inversion at constant CAS?

- a. TAS increases.
- b. Mach number increases.
- c. Mach number remains constant.
- d. TAS remains constant

64. What is V_{FE} ?

- a. The maximum speed for extending or retracting the flaps.

- b. The maximum speed for flight with the flaps extended.
- c. The maximum speed for the flight envelope.
- d. The minimum speed for extending or retracting the flaps.

65. What is V_{LO} ?

- a. The maximum speed with landing gear out.
- b. The maximum speed for retracting or extending the landing gear.
- c. The minimum speed for flight with landing gear out.
- d. The minimum speed for flight when operating the landing gear.

66. What does the green arc on an ASI indicate?

- a. V_{LE} at the lower end and V_{LO} at the upper end.
- b. V_{FE} at the lower end and V_{FO} at the upper end.
- c. V_{S1} at the lower end and M_{MO} at the upper end.
- d. V_{S1} at the lower end and V_{NO} at the upper end.

67. When, in flight, the needle of a needle and ball indicator is on the left and the ball on the right, the aircraft is?

- a. Turning left with too much bank.
- b. Turning right with not enough bank.
- c. Turning right with too much bank.
- d. Turning left with not enough bank.

68. Following 180° stabilized turn with a constant attitude and bank, the artificial horizon indicates?

- a. Too high pitch-up and too low banking.
- b. Too high pitch-up and correct banking.
- c. Attitude and banking correct.
- d. Too high pitch up and too high banking.

69. When standing at the South Pole in which direction will you be facing?

- a. North.
- b. South.
- c. East.
- d. West.

70. What is the approximate circumference of the earth?

- a. 5400 nm.
- b. 11800 nm.

- c. 21600 nm.
- d. 43400 nm.

71. An aircraft was over 'A' at 1435 hours flying direct to 'B'. Given : Distance 'A to 'B' 2900 NM, True airspeed 470 kt, Mean wind component 'out' +55 kt, Mean wind component 'back' -75 kt, The ETA for reaching the Point of Equal Time (PET) between 'A' and 'B' is?

- a. 1744.
- b. 1846.
- c. 1721.
- d. 1657.

72. What is the meaning of the term "standard time"?

- a. It is the time set by the legal authorities for a country or part of country.
- b. It is just another term for UTC.
- c. It is time related to the Prime meridian.
- d. It is constant at all points on the earth.

73. What is the local mean time, position 65°25'N 123°45'W at 2200 UTC?

- a. 0615.
- b. 1815.
- c. 1345.
- d. 1415.

74. The main reason that day and night, throughout the year, have different duration, is due to the?

- a. The changing distance between the Sun and the Earth.
- b. The changing speed of rotation of the earth.
- c. The changing shape of the orbit to the Earth around the Sun.
- d. The inclination of the ecliptic to the Equator.

75. Convert 95 meters/ second into knots?

- a. 135 kts.
- b. 155 kts.
- c. 185 kts.
- d. 166 kts.

76. If there is a 12 knot increase in headwind by what amount must the rate of descent be changed in order to maintain a 3° glideslope?

- a. It must be increased by 35 ft/ min.
- b. It must be decreased by 35 ft/ min.
- c. It must be decreased by 61 ft/ min.
- d. It must be increased by 61 ft/ min.

77. What will be the rate of descent when flying down a 7% glide slope, at a groundspeed of 540 knots?

- a. 3130 ft/min.
- b. 4830 ft/min.
- c. 3830 ft/min.
- d. 3740 ft/min.

78. An aircraft flying down a 3° ILS glideslope is at 35 nm DME from the threshold. Using the 1 in 60 rule and approximating 1 nm to 6000 ft, calculate the aircraft height above the runway threshold, assuming that the ILS glidepath crosses the threshold at a height of 50 ft?

- a. 6450 ft.
- b. 7455 ft.
- c. 10550 ft.
- d. 10015 ft.

79. Given: Position A 45° N, ?° E, Position B 45° N, 45° 15'E Distance A-B= 350 NM, B is to the East of A, What is the longitude of position A.

- a. 37° 00'E.
- b. 48° 00'E.
- c. 51° 15'E.
- d. 8° 15'E.

80. An increase of 0.15 mach results in an increase of 93 Kts TAS. The local speed of sound is?

- a. 560 Kts.
- b. 685 Kts.
- c. 620 Kts.
- d. 580 Kts.

81. Given the following: True track: 192° Magnetic variation: 7°E Drift angle: 5° left What is the magnetic heading required to maintain the given track?

- a. 190° M
- b. 192° M
- c. 197° M
- d. 188° M

82. Given the following: Magnetic heading : 060° Magnetic Variation: 8°W Drift angle: 4°right, what is the true track?

- a. 058°
- b. 054°
- c. 052°
- d. 056°

83. Given FL350, Mach 0.80, OAT -55°C. Calculate the values for TAS and local speed of sound (LSS)

- a. 460 kt, LSS 575 kt.
- b. 460 kt, LSS 570 kt
- c. 465 Kt, LSS 575 kt.
- d. 460 kt, LSS 567 kt.

84. 265 US-GAL equals? (Specific gravity 0.80)

- a. 803 kg.
- b. 810 kg.
- c. 800 kg
- d. 813 kg.

85. Given : TAS= 190 kt. True HDG= 085° ,W/V=110°(T)/ 50 kt, Calculate the drift angle and GS

- a. 8°R- 146 kt.
- b. 8°L- 146 kt.
- c. 8°L- 140 kt.
- d. 8°R- 140 kt.

86. Given: Runway direction 230° (T), Surface W/V 280° (T)/40kt. Calculated the effective cross-wind component?

- a. 31 kt
- b. 25 kts.
- c. 33 kts.
- d. 34 kts.

87. How do rhumb lines, (other than meridians), appear on Polar Stereographic charts?

- a. As straight lines.
- b. As lines concave to the nearer pole.
- c. As lines convex to the nearer pole.
- d. As ellipses around the pole.

88. Where is the convergency correct on a Transverse Mercator chart?

- a. At the datum meridian and the Equator.
- b. Only at the Equator and Poles.
- c. Only at the datum meridian.
- d. At the Parallel of Origin.

89. In which of the following projections will a plane surface touch the reduced earth at one of the Poles?

- a. Lambert's.
- b. Direct Mercator.
- c. Transverse Mercator.
- d. Stereographic.

90. Conversion of fuel volume to mass

- a) may be done by using standard fuel density values as specified in the Operations Manual, if the actual fuel density is not known.
- b) may be done by using standard fuel density values as specified in JAR - OPS 1.
- c) must be done by using actual measured fuel density values.
- d) must be done using fuel density values of 0.79 for JP 1 and 0.76 for JP 4 as specified in JAR

91. Standard masses may be used for the computation of mass values for baggage if the aeroplane

- a) is carrying 30 or more passengers.
- b) has 6 or more seats.
- c) has 30 or more seats.
- d) has 20 or more seats

92. Position A is 55° N 030° W and B is 54° N 020° W. What is the rhumb line bearing from A to B, if the great circle track from A to B, measured at A, is 100°T?

- a. 110°T.
- b. 284°T.
- c. 104°T.
- d. 090°T.

93. A revenue flight is to be made by a jet transport. The following are the aeroplane's structural limits:-Maximum Ramp Mass: 69 900 kg-Maximum Take Off Mass: 69 300 kg-Maximum Landing Mass: 58 900 kg-Maximum Zero Fuel Mass: 52 740 Kg Take Off and Landing mass are not performance limited .Dry Operating Mass: 34 930 kg Trip Fuel: 11 500 kg Taxi Fuel: 250 kg Contingency & final reserve fuel: 1 450 Kg Alternate Fuel: 1 350 kg The maximum traffic load that can be carried is:

- a) 21 170 kg
- b) 17 810 kg
- c) 21 070 kg
- d) 20 420 kg

94. Given: Chart scale is 1: 5 000 000. The chart distance between two points is 10 centimetres. Earth distance is approximately?

- a. 230 nm.
- b. 243 nm.
- c. 255 nm.
- d. 270 nm.

95. The total length of the 70° N parallel of latitude on a direct Mercator chart is 120 cm. What is the approximate scale of the chart at latitude 40°S?

- a. 1 : 16 050 000.
- b. 1 : 2 600 000.
- c. 1 : 7 250 000.
- d. 1 : 25 500 000.

96. A straight line drawn on a chart measures 4.63 cm and represents 125 NM. The chart scale is?

- a. 1 : 5 000 000.
- b. 1 : 6 000 000.
- c. 1 : 4 500 000.
- d. 1 : 6 500 000.

97. Two aircraft are flying eastwards around the earth. Aircraft A is flying along the 50N parallel of latitude and aircraft B is flying along the equator. If aircraft A is flying at a groundspeed of 340 knots, what would be the groundspeed of aircraft B both aircraft fly once round the Earth in the same journey time?

- a. 529 knots.
- b. 540 knots.
- c. 480 knots.
- d. 600 knots.

98. The sun rises at 50° N and 025°W on the 25th January at 0654 UTC. On the same day what time does it rise at 50° N and 040°E?

- a. 0254 UTC.
- b. 0154 UTC.
- c. 0234 UTC.
- d. 0514 UTC.

99. What is the longitude of a position 15 NM to the east of 58° 42'N 094° 00'W?

- a. 093° 51.3'W.
- b. 094° 13.0'W.
- c. 093° 31.1'W.
- d. 122° 31.1'W.

100. Total Air Temp is always... than Static Air Temp and the difference varies with...

- A) colder, altitude.
- B) warmer, altitude.
- C) warmer, TAS.
- D) colder, CAS.